Classification of Australian Native Forest Species Using Hyperspectral Remote Sensing and Machine-Learning Classification Algorithms

Xiao Shang and Laurie A. Chisholm

***Abstract*—Mapping forest species is highly relevant for many ecological and forestry applications. In Australia, the classification of native forest species using remote sensing data remains a particular challenge since there are many eucalyptus species that belong to the same genus and, thus, exhibit similar biophysical characteristics. This study assessed the potential of using hyperspectral remote sensing data and state-of-the-art machine-learning classification algorithms to classify Australian forest species at the leaf, canopy and community levels in Beecroft Peninsula, NSW, Australia. Spectral reflectance was acquired from an ASD spectrometer and airborne Hymap imagery for seven native forest species over an Australian eucalyptus forest. Three machine-learning classification algorithms: Support Vector Machine (SVM), AdaBoost and Random Forest (RF) were applied to classify the species. A comparative study was carried out between machine-learning classification algorithms and Linear Discriminant Analysis (LDA). The classification results show that all machine-leaning classification algorithms significantly improve the results produced by LDA. At the leaf level, RF achieved the best classification accuracy (94.7%), and SVM outperformed the other algorithms at both the canopy (84.5%) and community levels (75.5%). This study demonstrates that hyperspectral remote sensing and machine-learning classification has substantial potential for the classification of Australian native forest species.**

***Index Terms*—Forestry, remote sensing.**

I. Introduction

MAPPING forest species is important to effectively manage forests as a natural resource [1], monitor physiological stress [2]–[7], and investigate changes in species richness and diversity due to ecological disturbances such as bush fires and cyclones [8], [9]. Remote sensing is complementary to traditional field surveys for obtaining species information, particularly within large and inaccessible areas. Hyperspectral remote sensing facilitates the mapping of forest species due to finer spectral resolution and a more continuous spectrum than conventional multispectral remote sensing. Such characteristics have been demonstrated to detect subtle differences in chlorophyll, pigments and leaf structure, leading to an increased possibility of determining the characteristic spectral features for the analysis, classification and monitoring of forest species [10]–[12].

In Australia, the classification of native forest species using hyperspectral remote sensing data remains a challenge. Studies that have attempted to classify native forest species have not been entirely successful, nor have results been consistent. The ability to identify different kinds of eucalyptus species which belong to the same genus remains a difficult task [13]–[16]. Kumar and Skidmore [17] and Datt [18] have demonstrated that using spectrometers under laboratory conditions can effectively detect differences in spectral reflectance for several eucalyptus species. However, they also point out that not every species can be successfully differentiated from each other in a classification scheme, due to strong similarity in their spectral reflectance. Many researchers have used a number of classification algorithms, such as Maximum Likelihood Classification (MLC), Multiple Stepwise Discriminate Analysis (MDA) and Linear Discriminant Analysis (LDA) [10], [14], [19], to classify the spectral reflectance of different eucalyptus species. Limited success achieved, largely attributed to similarities in structural and chemical properties. Eucalyptus leaves are pendulous, forming an open canopy structure. Thus, the spectral reflectance obtained from remote sensing imagery is strongly affected by the background resulting in the high variation in spectral reflectance measured for the same species [10]. Furthermore, the spectral reflectance of different eucalyptus species belonging to the same genus has stronger similarities than those of species from different genera due to their similar biochemical components [14]. Overall, the result is that the classification algorithms utilized to date have not reliably detected such subtle differences.

In order to improve upon classification results, it has been recently suggested that the use of state-of-the-art classification algorithms and object-oriented classification techniques should be considered [20]–[23] as a means to overcome these limitations. Support Vector Machine (SVM), AdaBoost and Random Forest (RF) are three commonly used machine-learning classification algorithms which often provide higher classification accuracies for land-cover classification than previously achieved [24]–[26], yet their application to vegetation-related studies using hyperspectral remote sensing data is a relatively recent trend [27]. Increasing classification

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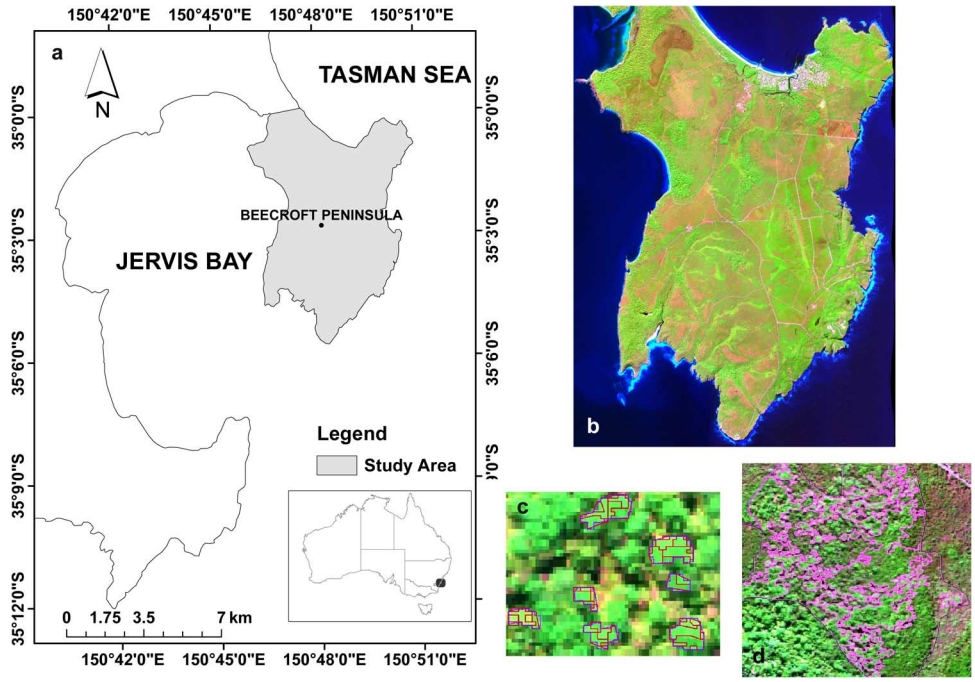

Fig. 1. (a) Study area of Beecroft Peninsula in southeastern Australia; (b) the HyMap imagery shown as the bands combination of band 79, 28, 14; (c) example of delineation (purple dash line) and segmentation (red solid line) of canopies; (d) example of delineation (black solid line) and segmentation (pink solid line) of vegetation communities.

accuracies with these algorithms suggest they may perform significantly better than conventional classification algorithms [24], [28]–[30] for vegetation studies. To date, these algorithms have not been widely applied to the mapping of forest species [31], [32].

The objective of this study was to investigate the potential of using machine-learning algorithms to classify several eucalyptus species at the leaf, canopy and community levels within a eucalyptus-dominated forest on the south coast of New South Wales. A leaf-level study was conducted using field spectroscopy based on the spectral reflectance of leaf samples collected from the field. In conjunction, canopy- and community-level studies were conducted based on airborne hyperspectral imagery where spectral reflectance was derived based on image objects representing the entire (or partial) canopy of individual tree or vegetation community patches in the imagery. This study focused on two specific aims: 1) to investigate which wavelengths indicate statistically significant differences between species at the leaf level, and 2) to evaluate three machine-learning algorithms—SVM, AdaBoost, and RF—at the leaf, canopy and vegetation community levels, compared with LDA classification, to assess whether these algorithms can effectively classify specific Australian native forest species.

II. Methodology

A. Study Area

The Beecroft Peninsula ($35°2'36.56''$S, $150°48'18.76''$E), is located on the south coast of NSW in southeastern Australia (Fig. 1) and is owned by the Department of Defence (DoD).

TABLE I
Studied Species and the Number of Measured/Extracted Spectral Reflectance Used in the Training and Testing for the Leaf- and Canopy-Levels Studies

Name	Scientific Name	Code	Leaf level		Canopy level		
			Train	Test	Train	Test	Number of Tree Crowns
Rough Barked Apple	*Angophora floribunda*	RBA	15	25	20	89	20
Grey Iron Bark	*E. paniculata*	GIB	15	25	25	101	25
Scribbly Gum	*E. sclerophlylla*	SG	15	25	25	105	25
Red Bloodwood	*E. gummifera*	RB	15	20	14	59	14
Silver-top Ash	*E. sieberi*	STA	15	25	25	106	25
Sydney Peppermint	*E. piperita*	SP	15	25	13	45	13
Bangalay	*E. botryoides*	BG	15	25	26	113	26
Total			105	170	148	618	148

Train — Number of training samples; Test — Number of testing samples

The vegetation on the Beecroft Peninsula is diverse and of high conservation value [33]. It is relatively undisturbed, because of the Peninsula's geographic isolation and the restrictions on access and development imposed by the DoD. The climate of the area is temperate-maritime, and thus characterized by mild temperatures throughout the year [34], with an average maximum temperature of 24°C during the summer and of 15°C during the winter. The average annual rainfall is close to 1250 mm distributed throughout the year, with the highest rainfall recorded during the austral autumn [34]. In this study, we conducted leaf- and canopy-level studies on six eucalyptus species and one non-eucalyptus species (Rough Barked Apple) commonly found on the Peninsula (Table I), and a community-level study on six common vegetation communities, which included four eucalyptus-dominated communities (Table II).

TABLE II
STUDIED VEGETATION COMMUNITIES AND THE NUMBER OF EXTRACTED SPECTRAL REFLECTANCE USED IN THE TRAINING AND TESTING FOR THE COMMUNITY-LEVEL STUDY

Name	Code	Training	Testing
Allocasuarina distyla Shrubland	AD	45	66
Banksia ericifolia Heath	BE	64	97
E. botryoides and *E. sclerophylla* Forest	BS	46	69
E. gummifera and *E. sieberi* Woodland	GS	51	77
E. paniculata and *Allocasuarina littoralis* Forest	PA	29	42
E. piperita and *Angophora floribunda* Forest	PPA	45	61
Total		280	412

B. Collection of Spectral Reflectance

1) Collection of Leaf-Level Spectral Reflectance Data: Leaf samples were collected in the field from 24 to 26 November 2010 using a .22BR rifle. An extendable pruning stick was used to collect the branches from trees less than 8 m in height. Two trees of each species were selected for sampling, and two branches were shot down from different aspects of the mid-upper crown of each tree. We collected 100 mature leaf samples for each species. The spectral reflectance of leaf samples was measured under laboratory conditions (i.e. dark room, $\sim 25°\mathrm{C}$), using an ASD FieldSpec Pro FR spectrometer (Analytical Spectral Devices, Boulder, CO, USA) which measures the spectral range from 350 nm to 2500 nm and records 2151 bands. The spectral resolution of the ASD FieldSpec FR is 3 nm for the region 350–1000 nm and 10 nm for the region 1000–2500 nm. Leaf samples were randomly spread on top of a flat black plate covering an area of approximately 15 cm × 10 cm and stacked six layers deep to ensure an infinite optical thickness [2]. Two 60 W halogen lamps were used as the light source at a 45° angle of incidence, focusing on the leaves at a distance of 50 cm to provide conditions of consistent illumination. The spectrometer was positioned at the nadir 25 cm above the plate centre with a 25° field of view (FOV), and configured to yield a spectral reflectance with an average over 25 spectra. Forty measurements were taken for each species (35 measurements for RB). The leaf samples were randomly shuffled after each measurement. The spectrometer was optimized with a white spectralon panel, exhibiting near-Lambertian properties, and the instrument was re-optimized with the panel after every five measurements. All measurements were conducted within four hours of the time of field collection to avoid water loss and changes in leaf properties.

2) Airborne Hyperspectral Imagery Acquisition and Pre-Processing: HyMap imagery (HyVista Corporation Pty Ltd, Castle Hill, NSW, Australia) was collected on 17 August 2008 at 11:00 am AEST, under clear-sky conditions. HyMap is an airborne hyperspectral sensor that captures 128 bands ranging from 446 nm to 2477 nm. Bandwidth intervals are 10 nm in the visible and near-infrared (NIR) wavelength, and 15–20 nm in the short-wave infrared (SWIR) [35]. The pixel size of this image is 2.8 m at 1200 m flying altitude. Our study area was covered by six image swaths, each swath being approximately 1.4 km wide (east–west), and 10–15 km long (north–south). The images were supplied by HyVista using atmospheric correction and Bidirectional Reflectance Distribution Function (BRDF) transformation. Georeferencing of the images was carried out in ENVI (Research Systems, Inc., Boulder, CO, USA) using the HyVista-supplied geocorrection file (overall RMS error ≈ 0.3 pixels). Beecroft Peninsula has mild temperatures throughout the year and is dominated by evergreen vegetation species. Thus, though the HyMap imagery was captured in the winter, the classification results would not be significantly affected by the seasonality.

3) Extraction of Canopy-Level Spectral Reflectance From HyMap Imagery: A GeoXH GPS unit (Trimble Navigation Limited, Westminster, CO, USA) was used to record tree locations. In some instances, a GPS signal from satellites could not be received because of the dense canopy cover; in these cases, we recorded the relative location (distance and bearing) of the tree from its two nearest GPS points. Each distance and bearing was recorded as the average of five readings. After post-processing using GPS Pathfinder (a program built by Trimble for differentially correcting the field data recorded by GeoXH), the maximum error in the GPS recordings ($\leq$75 cm) was much less than one pixel size (2.8 m), which is accurate enough to delineate canopy boundaries. The boundary of the tree canopy was delineated using ArcGIS by digitising tree canopy the for each of the sample trees, according to precise location data collected during fieldwork. Clark *et al.* [36] pointed out that tree species discrimination is best accomplished by object-oriented classification, thus each delineated tree canopy was segmented at a finer scale to generate several HyMap image objects using eCognition (Definiens Imaging, Munich, Germany). Segmentation is a region growing technique, and we carried out the segmentation based on the pixel values of all HyMap bands. The pixels included in the same image object have similar pixel values. Each of the image objects was assigned to the appropriate species class, based on *a priori* knowledge from fieldwork, and the mean reflectance values of each image object were extracted.

4) Extraction of Community-Level Spectral Reflectance From HyMap Imagery: The HyMap imagery and a digital (GIS) historical vegetation map [33] were used to obtain the spectral reflectance of each community patch. The historical map includes 27 vegetation communities produced by Skelton and Adam [33] in 1994 while undertaking a vegetation survey for DoD. The applicability of the historical vegetation map was evaluated by collecting ground-truth data during June 2011. A field evaluation supported by observations from DoD staff showed no significant change in the spatial distribution of vegetation communities over this period, attributed to little disturbance activity. Initially, the HyMap imagery was overlaid by the historical vegetation map to locate vegetation communities [33]. The range of each community was refined by visual interpretation of the HyMap imagery, and each of the target community patches was delineated using ArcGIS. Each community patch was then converted to several homogenous HyMap image objects through segmentation in eCognition. Each of the image objects was assigned to an appropriate class by its dominant species based on *a priori* knowledge from the historical vegetation map, and mean reflectance values of each image object were calculated and extracted.

C. Machine-Learning Algorithms

Three machine-learning algorithms, Support Vector Machine (SVM), AdaBoost and Random Forest (RF), commonly used in remote sensing land-cover classification, were assessed in this study for classification of certain Australian native forest species. The machine-learning classification results were compared with LDA classification which has been extensively used in vegetation studies [36].

The supervised machine-learning method, SVM, based on statistical learning theory, was originally developed to solve the binary classification problem [37]. The SVM classification separates the classes with a decision surface (or a set of decision surfaces). This decision surface, called hyperplane, maximizes the functional margin between classes, while minimizing the training error [38]. For multiclass problems, SVM is to convert the single multiclass problem into multiple binary classification problems [39]. In addition, kernel functions are commonly used to construct a nonlinear hyperplane. Here, we used radial basis function (RBF) kernel selected using 10-fold cross-validation test (see Section II-E).

AdaBoost [40] is a supervised machine-learning method. It determines classification rules from samples by constructing a "strong" classifier through the combination of many "weak" classifiers [41]. Initially, all of the samples are equally weighted, but for the following iteration, the weights which indicate the importance of samples in the classification are individually modified [42]. The weights of the training observations misclassified by the classifier in previous rounds are increased. Conversely, the weights will be decreased for samples that were correctly classified [42]. Therefore, AdaBoost improves classifier performance by focusing on the data that was misclassified in the previous classification iteration rounds [41].

RF [43] is a tree-based machine-learning classification algorithm [44]. RF classification uses an ensemble classification method which consists of a set of decision tree classifiers, and the improved bootstrap method [45]. The input vectors are randomly sampled for each of the tree classifiers independently, and each tree casts a unit vote for the most popular class to classify the input vector. Specifically, for each of Ntrees iterations (Ntrees is the number of trees to build), a new bootstrap sample from the training set will be selected to build an un-pruned tree onto the bootstrap. At each internal node, mtry predictors (number of predictors sampled for splitting at node) are randomly selected, and used to determine the best split. After all of the iteration steps, RF outputs the majority vote from all individually trained trees.

Linear Discriminant Analysis (LDA), developed in 1936 by Fisher, is an effective technique for data classification. LDA has been extensively used in statistics, pattern recognition, and machine-learning to find a linear combination of features that characterizes or separates two or more classes of objects [41]. LDA maximizes the ratio of variance between the classes, to the variance within the classes, to guarantee the maximum separation, and it subsequently generates a decision surface between the classes [46].

D. Spectral Discriminative Analysis

Analysis of variance (ANOVA) is a statistical method to assess statistically significant differences between two or more sample groups. It has been widely used in species discrimination studies to identify the wavelengths that show statistically significant differences between species [16], [47]–[49]. For studies of species discrimination, a Tukey-Kramer test [50] is commonly used after an ANOVA test to determine which species pairs are responsible for a significant result. In order to identify statistically different spectral wavelengths for species separation, a one-way ANOVA and a post-hoc Tukey-Kramer test were carried out at the leaf level in present study. There were 21 combinations based on our seven tree species. Therefore, if a wavelength can be effectively used to distinguish between all the species, its frequency (the number of the combinations showing significant spectral differences at certain wavelength for pair-wise comparisons) would be 21.

E. Classification

The three machine-learning classification algorithms, SVM, AdaBoost and RF, were applied at the leaf, canopy and community levels. In order to assess their performance in the classification of forest species, the comparison were carried out among these algorithms at each level. Their classification results were also compared with the LDA classification method commonly used in the classification of forest species [12], [19], [36], [51] to assess whether these machine-learning algorithms can produce a better results than a traditional classification method. At each level, approximately 40% of measured/extracted spectral reflectance data were randomly selected for each species as a training set and the other spectral reflectance data were retained as the testing set. The sizes of training and testing sets in the leaf-, canopy-, and community-level studies were summarized in the Table I and Table II.

Parameter tuning for machine-learning algorithms used a 10-fold cross-validation, implemented with the range of parameter values from 100 to 1500 $(\text{interval} = 100)$ for parameters C (the parameter C controls the trade-off between errors of the SVM on training data and margin maximization) in SVM, Ntrees in RF and iterations in AdaBoost; and range from 0.1 to 2 $(\text{interval} = 0.1)$ for parameter γ (parameter γ controls the shape of the separating hyperplane) in SVM. The SVM was carried out using an RBF kernel in an open source package LIBSVM [52], with two user-defined parameters: $\text{C} = 1000$ and $\gamma = 1$. AdaBoost and RF were carried out using the R packages 'adabag' and 'randomForest', respectively. AdaBoost was conducted using 500 iterations. RF has two user-defined parameters: Ntrees and mtry [43]. The algorithm is not sensitive to mtry and typically $\text{mtry} = \sqrt{k}$ (k = number of bands) performs reasonably well [45]. We set $\text{mtry} = \sqrt{128}$, and $\text{Ntrees} = 1000$ from 10-fold cross-validation test.

As the studied algorithms require training and testing samples for classification, the samples were split into a training set and a testing set. At each level, approximately 40 per cent of measured/extracted spectral reflectance was randomly selected for each species as a training set, with the remaining spectral reflectance retained as the testing set.

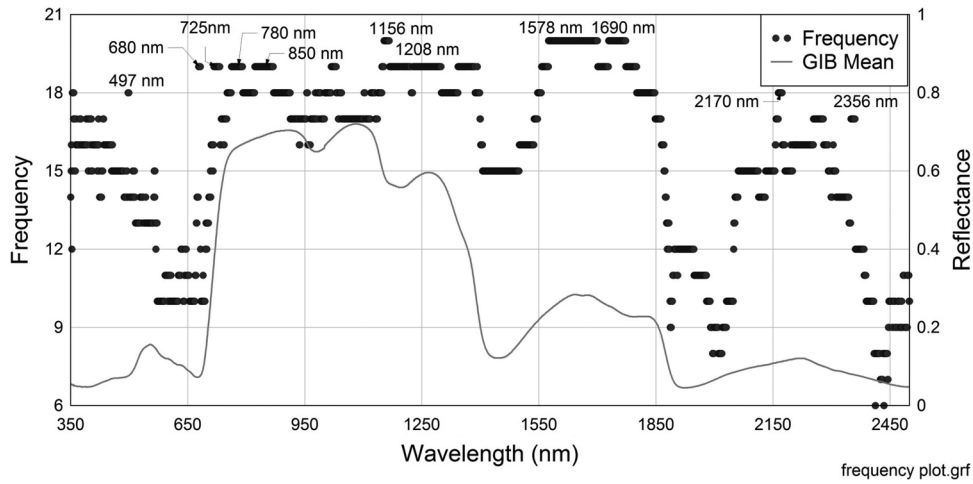

Fig. 2. Frequency plot of statistically significant differences, obtained using the Tukey-Kramer test between spectral reflectance at every band for all the species at the leaf level.

III. Results

Fig. 2 shows that there was no single wavelength that can be used to discriminate all the species at the leaf level because the highest frequency value was 20, which provided by spectral regions close to wavelengths of 1156 nm, 1578 nm, and 1690 nm. The mean reflectance curve of GIB (Grey Iron Bark) is overlain on the frequency plots for the better interpretation. This was followed by 19 discriminations with spectral regions at approximate wavelengths of 497 nm, 680 nm, 725 nm, 780 nm, 850 nm, and 1208 nm. Many spectral regions can support 18 discriminations among species, which spread over the full range of the spectrum. It was observed that the majority of the wavelengths (1537 out of 2151) can support 15 pair-wise discriminations or more. The lowest frequency value occurred in the SWIR region, near wavelengths of 2010 nm and 2410 nm.

At the leaf level, the three algorithms achieved very good classification accuracies (Table III). Table III indicates that all of the machine-learning classification algorithms significantly improved the overall accuracy delivered by LDA (69.4%). The best overall accuracy (94.7%) was achieved by RF with the highest Kappa coefficient value at 0.938. In particular, most producer's and user's accuracies produced by RF and AdaBoost exceeded 80%. SG and STA can be robustly classified from the other species. The remaining species did not produce consistent results between algorithms. SVM achieved a high producer's accuracy for BG and RB, whereas AdaBoost and RF delivered relatively low accuracy. Conversely, AdaBoost and RF produced a high producer's accuracy for RBA and STA, but SVM achieved relatively low accuracy. Despite these inconsistencies, all of the machine-learning classification algorithms significantly improved the producer's accuracy delivered by LDA, and a high level of user's accuracy (>80%) was also achieved by AdaBoost and RF for all of the species.

At the canopy level, the best overall accuracy was produced by SVM (84.5%) (Table IV). All of the machine-learning classification algorithms performed better, in terms of overall accuracy, than LDA (65.4%). The best Kappa coefficient (0.816) was produced by SVM, whereas LDA delivered the lowest Kappa coefficient of 0.588. SVM, AdaBoost and RF could consistently produce a high producer's accuracy for RBA, RB, GIB and STA, and delivered a relatively lower producer's accuracy for SP. Table IV shows that all the machine-learning classification algorithms produced a producer's accuracy of over 70% for all of the species included in the study, and significantly outperformed the producer's accuracy delivered by LDA. SVM achieved a high user's accuracy at the canopy level (83.7%): a 15% improvement over the user's

TABLE III
Accuracy Assessment Results for the Classifications Achieved Using AdaBoost, RF, SVM, and LDA at the Leaf Level

	AdaBoost		RF		SVM		LDA	
	Pac	Uac	Pac	Uac	Pac	Uac	Pac	Uac
RBA	96.0	80.0	96.0	82.8	80.0	90.9	68.0	68.0
BG	80.0	95.2	80.0	95.2	92.0	76.7	88.0	53.7
RB	75.0	88.2	85.0	100.0	90.0	66.7	50.0	76.9
GIB	88.0	88.0	100.0	92.6	84.0	87.5	84.0	75.0
SP	96.0	96.0	100.0	96.2	96.0	100.0	56.0	66.7
SG	100.0	96.2	100.0	100.0	100.0	100.0	88.0	88.0
STA	100.0	96.2	100.0	100.0	72.0	100.0	48.0	70.6
Average	**90.7**	**91.4**	**94.4**	**95.3**	**87.7**	**88.8**	**68.9**	**71.3**
OA	**91.2**		**94.7**		**87.6**		**69.4**	
Kappa	**0.897**		**0.938**		**0.856**		**0.643**	

Pac — Producer's accuracy (%); Uac — User's accuracy (%);
OA — overall accuracy (%);

TABLE IV
Accuracy Assessment Results for the Classifications Achieved Using AdaBoost, RF, SVM, and LDA at the Canopy Level

	AdaBoost		RF		SVM		LDA	
	Pac	Uac	Pac	Uac	Pac	Uac	Pac	Uac
RBA	83.1	86.0	79.8	78.9	82.0	84.9	68.5	57.5
BG	74.3	75.0	84.1	82.6	85.0	85.0	80.5	61.9
RB	84.7	70.4	81.4	82.8	79.7	81.0	45.8	69.2
GIB	96.0	86.6	82.2	76.1	94.1	81.9	66.3	63.8
SP	73.3	86.8	77.8	81.4	77.8	76.1	48.9	84.6
SG	72.4	75.2	76.2	78.4	81.0	84.2	51.4	71.1
STA	84.0	90.8	80.2	84.2	85.8	92.9	77.4	68.9
Average	**81.1**	**81.5**	**80.2**	**80.6**	**83.6**	**83.7**	**62.7**	**68.1**
OA	**81.4**		**80.4**		**84.5**		**65.4**	
Kappa	**0.780**		**0.769**		**0.816**		**0.588**	

Pac — Producer's accuracy (%); Uac— User's accuracy (%);
OA — overall accuracy (%);

TABLE V
ACCURACY ASSESSMENT RESULTS FOR THE CLASSIFICATIONS ACHIEVED USING ADABOOST, RF, SVM, AND LDA AT THE COMMUNITY LEVEL

	AdaBoost		RF		SVM		LDA	
	Pac	Uac	Pac	Uac	Pac	Uac	Pac	Uac
AD	75.8	79.4	71.2	79.7	77.3	86.4	68.2	63.4
BE	86.6	85.7	85.6	85.6	85.6	90.2	79.4	76.2
BS	85.5	73.8	79.7	64.7	78.3	78.3	71.0	72.1
GS	92.2	62.8	87.0	62.0	84.4	65.7	55.8	64.2
PA	28.6	70.6	33.3	58.3	35.7	45.5	47.6	55.6
PPA	54.1	80.5	50.8	79.5	70.5	71.7	67.2	59.4
Average	**70.5**	**75.5**	**67.9**	**71.6**	**72.0**	**73.0**	**64.9**	**65.2**
OA	**75.0**		**72.1**		**75.5**		**66.7**	
Kappa	**0.694**		**0.659**		**0.702**		**0.596**	

Pac — Producer's accuracy (%); Uac — User's accuracy (%);
OA — overall accuracy (%);

accuracy produced by LDA. AdaBoost and RF also produced user's accuracies that exceeded 80%. User's accuracy showed inconsistent results for each of the species delivered by different classification algorithms, mainly because user's accuracy correlated to the number of misclassified species. However, all the machine-learning classification algorithms produced reasonable user's accuracy for each of the species. In particular, SVM and RF produced a user's accuracy of above 76% for all species.

The classification accuracy from all the classification algorithms at the community level is summarized in Table V. The best overall accuracy (75.5%) was produced by SVM with the highest Kappa coefficient value at 0.702, and the lowest overall accuracy (66.7%) and Kappa coefficient value (0.596) was produced by LDA. The classes of BE and GS had the highest producer's accuracy. AdaBoost delivered the best producer's accuracy for BE (86.6%) and GS (92.2%). All machine-learning algorithms could produce both producer's and user's accuracies for the AD class of over 70%. For the PPA class, SVM delivered a producer's accuracy of 70.5% and user's accuracy of 71.7%, and both AdaBoost and RF produced a low producer's accuracy and a relatively high user's accuracy. In addition, the PA class had a high commission error for all machine-learning algorithms, resulting in a low producer's accuracy for AdaBoost (28.6%), RF (33.3%) and SVM (35.7%). Closer examination showed that most of these errors of commission arose because PA was significantly misclassified to GS. Therefore, GS had a high producer's accuracy for all machine-learning algorithms, but a relatively low user's accuracy caused by high omission errors.

IV. DISCUSSION

A. Classification of Australian Native Forest Species at Different Levels

In this study, we found that wavelengths of 497 nm, 680 nm, 725 nm, 780 nm, 850 nm, 1578 nm, 1690 nm, 2170 nm, and 2356 nm are the most useful bands for the discrimination of the eucalyptus species studied. With respect to the wavelengths of 497 nm and 680 nm, within the visible chlorophyll absorption regions, Goodwin *et al.* [14] demonstrated that amalgamated groups of eucalyptus species (E.saligna, E.pilularis and E.paniculata) can be discriminated from S.glomulifera and mesic vegetation (primarily rainforest species) at the 680 nm wavelength. We have extended this finding to within-genus species discrimination with our results showing statistically significant interspecies differences at the wavelength of 680 nm, consistent with the findings of Goodwin [14]. The 725 nm wavelength located in the red-edge region, known to be closely associated to foliar chlorophyll content [14], [16], [53], [54], also showed statistically significant differences among eucalyptus species—a consistent finding among similar studies where foliar chemistry is a key discriminating factor [14], [16], [53], [54]. The significant differences at the 850 nm wavelength, which has a strong correlation with chlorophyll content [55], is consistent with Datt [56] for discrimination of eucalyptus, and the subsequent development of two spectral indices for eucalypt discrimination [55], [57]. As noted by Kumar [16], our study found significant differences in SWIR among the eucalyptus species, however, we have isolated the differences to particular wavelengths of 1578 nm, 1690 nm, 2170 nm, and 2356 nm. It is likely these are attributed to differences in foliar biochemical features associated with absorption regions of lignin, protein, or nitrogen. Eucalyptus species also showed statistically significant differences at the water absorption spectral regions of 1046 nm, 1156 nm, 1208 nm, 1429 nm, and 1971 nm; however, these wavelengths are very sensitive to leaf water content and cannot be used to discriminate between species [15].

In general, the classification accuracy decreases from leaf level to community level with the decrease of spectral purity. Leaf-level study to classify the various eucalyptus species is one of the most important prerequisites to the future use of airborne-hyperspectral sensors for mapping eucalyptus species in large spatial extents. Our overall accuracy level was higher than the 72.45% observed by Datt [19] in the classification of eucalyptus based on laboratory-measured leaf spectral reflectance data and LDA. This significant improvement was due to the advanced machine-learning algorithms, coupled with ideal laboratory conditions and sufficient spectral information. The best classification result achieved in this study was similar to the 96% accuracy produced by Pu and Liu [58] using segmented Canonical Discriminant Analysis (CDA) and LDA to classify 13 tree species, including two conifers and 11 broadleaf species. However, LDA produced a poor overall classification accuracy for eucalyptus species. Considering that eucalyptus species which belong to the same genus have a strong similarity in their spectral reflectance [13]–[15], we therefore conclude that improved classification of eucalyptus species is possible and that the accuracy achieved by state-of-the-art machine-learning algorithms is approaching that for northern hemispheric forest species with relatively more easily distinguishable spectral properties.

At the canopy level, although a previous attempt [14] failed to classify eucalyptus species from hyperspectral remote sensing imagery, the machine-learning classification algorithms employed in this study achieved a high level of accuracy in distinguishing between certain eucalyptus species. The maximum accuracy achieved was 84.5%, using the SVM classification. This result is higher than the overall accuracy (66.6%) for spectral-based classification produced by Sensie *et al.* [29], also using SVM, to classify 11 rainforest types,

and is close to the 86.7% accuracy using RF and including a Normalized Difference Vegetation Index (NDVI) and digital elevation data. Our results can be compared to a study by Banskota *et al.* [1], who focused on classifying within-genus forest species by applying wavelets transformation to airborne hyperspectral data (AVIRIS) and by using LDA to classify three pine species (Pinus spp.) achieving 74.2% accuracy. Our overall classification accuracy, achieved by machine-learning classification, ranged from 80.4% to 84.5%. Conversely, our LDA classification accuracy was lower than the 66.7% achieved by Banskota *et al.* [1] using spectral-based LDA classification. This comparison further confirms the benefits of using machine-learning algorithms for the classification of forest species. Our classification results are comparable with the study conducted by Bunting and Lucas [59], which classified a number of Australian forest species including a few eucalyptus species, achieving a result of 87% accuracy in mapping delineated tree crowns via Multiple Stepwise Discriminate Analysis (MDA); the results were also similar to a study by Youngentob *et al.* [60] using multiple endmember spectral mixture analysis (MESMA) and continuum removal to classify eucalyptus subgenera, with an overall accuracy rate of 83%.

The classification accuracy achieved at the community level was lower than that at the canopy level, probably due to the inclusion of a greater number of pixels consisting of several different classes (e.g., understory vegetation and/or bare soil). Specifically, non-eucalyptus dominated communities consistently yielded a relatively high producer's and user's accuracy, demonstrating that classification can distinguish them from eucalyptus-dominated communities, and also that a high level of mapping accuracy can be achieved as a result of their strong spectral differences from the eucalyptus-dominated communities. For eucalyptus-dominated communities, although a promising overall classification accuracy was achieved, there was confusion between some communities. In particular, the poor classification accuracy of the PA class indicated that it cannot be classified separately from other vegetation communities, especially from GS. This may be due to the inherent characteristics of the PA community, which comprises both eucalyptus and allocasuarina. Eucalyptus with pendulous leaves mixed with allocasuarina with needle-shaped leaves form an extremely open canopy structure, therefore, the PA community has a higher proportion of open canopy than a eucalyptus-dominated community. Previous studies have concluded that forest species with an open canopy structure yield a lower classification accuracy than those with a closed canopy, because the open canopy appears semi-transparent in remote sensing imagery; understory vegetation species and soil therefore significantly affect spectral reflectance [14], [61].

B. Comparison of the Machine-Learning Classification Algorithms

At all levels, the machine-learning algorithms outperformed the LDA classification. However, the classification results were inconsistent between the levels. At the leaf level, RF outperformed the other algorithms, and SVM can produce the best classification accuracy at the canopy and community levels. This is likely because RF has outstanding performance when the number of variables (number of spectral bands) is much larger than the number of observations (number of samples) [62], [63]. Therefore, spectral reflectance obtained for leaf-level study with more than 2000 spectral bands adequately meets the high performance criteria for RF when compared to HyMap, which has 128 spectral bands. The comparative study conducted at the canopy and community levels among SVM, AdaBoost and RF showed consistent results. SVM achieved the best classification accuracy for both canopy and community levels, followed by AdaBoost and RF. According to the producer's and user's accuracy, all of these machine-learning classification algorithms showed a strong capability of classifying eucalyptus species. SVM produced the best producer's accuracy and user's accuracy. In particular, the high user's accuracy achieved by SVM for all of the species at the canopy level suggests that it can be used to map eucalyptus species effectively with HyMap imagery.

Additionally, in terms of our observations during the classification process, SVM and RF classifications were much faster than AdaBoost which can become slow when handling many bands on large training sets [24]. This is in agreement with previous studies that have demonstrated that SVM and RF are more computationally efficient than AdaBoost [24], [25]. SVM does not use the entire set of training samples, because only the training samples that lie on the margin (Support Vectors) are able to contribute to the construction of the hyperplane [39]. RF selects only a subset of bands for training, while AdaBoost trains the algorithm using entire bands of training samples [24], so RF can handle significantly larger problems before slowing. Furthermore, RF has fewer user-defined parameters and provides better interpretation than SVM, because the algorithm can measure the importance of the bands to classification accuracy [29]. However, the differences between Kappa coefficients suggested that the SVM results in a more reliable method than either AdaBoost or RF for mapping certain Australian native forest species at the canopy and community levels.

C. Potential for Australian Native Forest Species Natural Resource Management

In this study, classification has been comprehensively investigated at three different levels: from the ideal leaf level (a laboratory-based study) to realistic image-based canopy- and community-level studies, the findings of which are of value in terms of their applicability. The best results observed at each level were encouraging. They confirmed that it is not only possible to discriminate between eucalyptus species in theory, but also that discrimination and mapping using airborne hyperspectral imagery are effective and applicable. The techniques evaluated in this study are practically feasible in the mapping of forest species over large spatial extents. The study confirmed the effectiveness and robustness of SVM and RF as well as their good performance in handling hyperspectral data. This study also confirmed that machine-learning algorithms have a strong capability for classifying forest species with an open canopy structure in practice. The success achieved in this study provides a robust method to conduct within-genus species discrimination. The limitations of this study were: 1) the classification was

based on the spectral information only; and 2) airborne-based hyperspectral imagery is relatively expensive. In order to overcome the first limitations, further studies are currently being undertaken to improve the current classification results by: 1) increasing the ability to discriminate of the species by using complementary texture measures; and 2) reducing the spectral variation within a species by applying data transformation methods. In addition, with the advances in technology, hyperspectral imagery may become more affordable in the near future. The outcomes from this study therefore, have major implications for the mapping, assessment and conservation of forests and the diversity of forest species.

V. Conclusion

This study has demonstrated that the application of advanced machine-learning classification algorithms to hyperspectral remote sensing data can effectively classify between specific Australian native forest species at the leaf, canopy, and community levels. Previous studies [13], [14] suggested that eucalyptus species are difficult to classify from each other using hyperspectral remote sensing imagery. This study however delivered encouraging results for the use of hyperspectral imagery to classify some eucalyptus species at each level (Leaf level 94.7%; Canopy level 84.5%; Community level 75.5%). Three machine-learning classification algorithms, SVM, AdaBoost, and RF, were evaluated and compared with the LDA classification algorithm. All of these classification algorithms significantly improved the LDA classification result; RF produced the best classification result for the leaf-level study, and SVM achieved the best classification result at both canopy and community levels. Therefore, RF and SVM were found to be the optimal algorithms for classification of certain eucalyptus species. In summary, the use of hyperspectral data and machine-learning algorithms has a great potential to map certain eucalyptus species. This study highlighted the potential for adoption of these methods by industry with respect of Australian native forest species, particularly eucalyptus species. This could lead to improvements in environmental sustainability through better management and protection of native vegetation communities.

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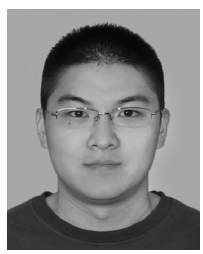

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